Practice 1: Train a deep model to classify lung diseases

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Abstract

Large-scale chest X-ray classification is investigated using two deep learning architectures, Swin Transformer Base (Swin-B) and ConvNeXt, trained on the ChestX-ray14 dataset. All images are resized to 224×224 *pixels, and the task is formulated as multi-label classification over 14 thoracic disease categories. Model performance is evaluated using micro- and macro-averaged area under the ROC curve (AUC). To ensure robustness, each experiment is repeated ten times, and the mean and standard deviation of AUC are reported.*

Experimental results indicate that directly training Swin-B with a softmax-based formulation leads to near-chance performance, underscoring the importance of an appropriate multi-label objective. Replacing the cross-entropy loss with binary cross-entropy with logits and adopting sigmoid-based probability estimation substantially improves convergence. In addition, ConvNeXt fine-tuned from ImageNet consistently outperforms training from scratch, achieving higher AUC and lower variance, particularly for rare disease categories. These findings highlight the critical role of loss design and transfer learning in large-scale medical image classification.

1. Introduction

In the medical domain, large volumes of imaging data require expert interpretation to identify subtle or latent pathological patterns. Manual diagnosis is time-consuming and may lead to diagnostic errors, particularly false negatives. To address these challenges, machine learning techniques have been increasingly adopted to assist clinicians by improving diagnostic accuracy, reducing false negative rates, and increasing true positive detection. Training robust models on large and diverse datasets is critical for enabling the discovery of disease patterns that may be difficult to identify through manual inspection.

This study investigates two deep learning architectures, Swin Transformer Base (Swin-B) and ConvNeXt, for large-scale chest X-ray classification on the ChestX-ray14 dataset. All images are preprocessed to a uniform resolution of 224×224 pixels and converted to single-channel grayscale format. The task is formulated as multi-label classification to account for the presence of multiple co-occurring thoracic diseases.

2. Dataset

The ChestX-ray14 dataset contains 112,120 frontal chest radiographs annotated with 13 thoracic disease labels and an additional No Finding label. Following common practice, 25,596 images are reserved for testing, while the remaining images are split into 75,312 training images and 11,212 validation images.

All images are resized to 224×224 pixels and normalized using ImageNet statistics. During training, data augmentation is applied in the form of random horizontal flipping and small random rotations. No augmentation is applied during validation or testing. All splits follow the official patient-independent partitioning to avoid data leakage.

3. Swin-Base Architecture

A Swin Transformer Base–style backbone is adopted, featuring a hierarchical window-based self-attention design. The network begins with a patch embedding layer implemented as a 4×4 convolution with an embedding dimension of 128. This is followed by four Swin Transformer stages with depths $(2, 2, 18, 2)$. Patch merging layers are applied between stages to progressively reduce spatial resolution while increasing channel dimensionality from 128 to 1024.

Each stage consists of window-based multi-head self-attention and multilayer perceptron (MLP) blocks with residual connections and stochastic depth. A customized classification head is added on top of the backbone, including an ensemble prediction layer that outputs logits for 25 target classes.

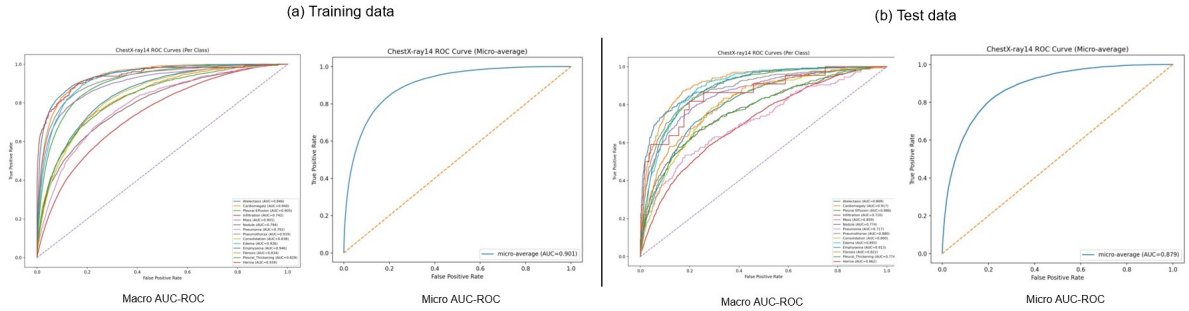

Figure 1. Training and test ROC curves for ImageNet pretraining, showing micro- and macro-averaged AUC at the best epoch.

4. ConvNeXt Architecture

ConvNeXt is adopted as the backbone network for chest radiograph analysis. ConvNeXt is a modernized convolutional architecture that incorporates design principles inspired by Vision Transformers while retaining the inductive biases of convolutional neural networks. The architecture follows a hierarchical design composed of multiple stages with progressively increasing channel dimensions and decreasing spatial resolutions.

Each stage consists of repeated ConvNeXt blocks. A ConvNeXt block contains a depthwise convolution with a large kernel size (7×7) to efficiently enlarge the receptive field, followed by Layer Normalization applied in a channels-last format. The normalized features are passed through a pointwise expansion layer, a GELU non-linearity, and a pointwise projection layer. Residual connections are employed to facilitate gradient flow and improve optimization stability.

The model is optimized using binary cross-entropy loss with logits and the AdamW optimizer, with an initial learning rate of 3×10^{-4} and a weight decay of 0.05. A batch size of 32 is used, and training is conducted for a fixed number of epochs. The checkpoint with the lowest validation loss is selected for evaluation.

5. Experiment

The classification task is formulated as a multi-label problem with $C = 14$ disease categories. Given an input image x, the ConvNeXt backbone produces a feature representation $f(x)$. The original ImageNet classifier is replaced with a linear layer that outputs C logits:

$$\mathbf{z} = W f(x) + b, \quad \mathbf{z} \in \mathbb{R}^{14}. \tag{1}$$

Independent sigmoid activations are applied to obtain class-wise probabilities:

$$\hat{\mathbf{y}} = \sigma(\mathbf{z}). \tag{2}$$

5.1. Swin-Base Architecture

Initial experiments using Swin-B with a softmax-based cross-entropy formulation result in near-chance performance, with an average training AUC of approximately 0.54. This behavior indicates that the model fails to learn meaningful decision boundaries under an incorrect single-label assumption.

To address this issue, the loss function is replaced with binary cross-entropy with logits, and sigmoid-based probability estimation is employed instead of argmax-based prediction. This modification aligns the training objective with the multi-label nature of the dataset and enables stable convergence.

5.2. ConvNeXt Architecture

Model performance is evaluated using both micro- and macro-averaged AUC. Micro-AUC aggregates predictions across all disease categories and reflects overall performance, while macro-AUC averages per-class AUCs and evaluates balanced performance across diseases.

Fine-tuning from ImageNet: ConvNeXt is initialized with ImageNet-pretrained weights and fine-tuned on the ChestX-ray14 training set. The best-performing checkpoint is obtained at epoch 7 based on validation loss.

Training from Scratch: ConvNeXt is trained without large-scale pretraining using the ChestX-ray14 training set.

As shown in Figure 1, ImageNet-pretrained models achieve higher ROC-AUC values across both training and test sets, with micro-AUC values approaching 0.90. The relatively small gap between training and test performance indicates good generalization and limited overfitting.

In contrast (Table 1 and Table 2), models trained from scratch converge more slowly and exhibit lower ROC-AUC values across most disease categories. Performance degradation is particularly pronounced for rare pathologies such as Mass and Infiltration, highlighting the difficulty of learning robust representations without pretraining.

Statistical hypothesis testing further supports these observations. The null hypothesis assumes no performance difference between training from scratch and ImageNet fine-tuning. Welch's two-sample t-test is applied to test-set AUC values across repeated runs. The null hypothesis is rejected for both micro- and macro-averaged AUC, indicating a statistically significant (0.01) improvement obtained

through ImageNet pretraining ($p < 0.001$).

Pretraining	Micro AUC	Macro AUC
Scratch	0.864 ± 0.0121	0.792 ± 0.0154
ImageNet	0.901 ± 0.0098	0.866 ± 0.0063

Table 1. Comparison of micro- and macro-averaged AUC for models trained on training data from scratch and with ImageNet pretraining.

Pretraining	Micro AUC	Macro AUC
Scratch	0.834 ± 0.0079	0.723 ± 0.0086
ImageNet	0.879 ± 0.0145	0.830 ± 0.0124

Table 2. Comparison of micro- and macro-averaged AUC for models trained on test data from scratch and with ImageNet pretraining.

5.3. Summary

Experimental results indicate that Swin-B is highly sensitive to the choice of loss formulation and probability estimation. Employing binary cross-entropy with logits and sigmoid-based predictions is essential for achieving meaningful performance in multi-label chest X-ray classification.

ConvNeXt fine-tuned from ImageNet consistently outperforms training from scratch, achieving higher AUC and lower variance across both common and rare thoracic diseases. These findings underscore the importance of transfer learning for medical image analysis, even in the presence of large-scale labeled datasets.

Furthermore, training a single backbone across multiple chest X-ray datasets can be effectively achieved by aligning heterogeneous annotations into a unified label space and applying masked multi-label supervision.